Module Review: APS-dependent Assimilatory Sulfate Reduction in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI txid 160488; proteome UP000000556) **Module scope:** sulfate → APS → sulfite → sulfide (direct APS branch) **Local bucket reviewed:** KEGG `ppu00920` "Sulfur metabolism" (54 candidate genes)

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1. Executive summary

The APS-dependent assimilatory sulfate-reduction module is **satisfiable and essentially complete** in *P. putida* KT2440. All three catalytic steps map to real, confidently annotated KT2440 genes:

1. **Sulfate activation → APS:** ATP sulfurylase **CysD (PP_1303)** + **CysNC (PP_1304)**.
2. **APS reduction → sulfite: CysH (PP_2328)**, UniProt-annotated *adenosine 5'-phosphosulfate reductase*, **EC 1.8.4.10** (thioredoxin-dependent, 4Fe-4S). This is the APS-reducing (plant/cyanobacterial-type) enzyme, confirming the module's **direct-APS scope** rather than the PAPS-reductase route.
3. **Sulfite reduction → sulfide:** siroheme hemoprotein **CysI (PP_2371)**, fed by the **ferredoxin/FprA route**.

Two curation-critical points emerge from species-specific evidence:

- The sulfite-reduction step in *P. putida* operates via the **Ferredoxin/Fpr-fed CysI route**, with **FprA** = **PP_1638** (adjacent to its regulator *finR* = PP_1637) supplying electrons. FprA is **missing from the candidate metadata** — a genuine electron-supply gap. The CysJ/CysI NADPH architecture is **not** the operational route here.
- The bucket is dominated (~30/54) by **organosulfur-scavenging and cysteine/methionine-synthesis genes that lie outside the module boundary**, and it contains at least one **over-propagated annotation (PP_1703)** that is actually a molybdopterin nitrate reductase.

Evidence transfer is **strong**: the key phenotypic study (P 23794620) uses *P. putida* strains (DSM 3601, and references S-313) and cites KT2440 locus tags (finR = PP1637), so conclusions transfer directly to KT2440.

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2. Target-organism pathway definition

Included (module boundary): the two-electron activation of inorganic sulfate to APS (ATP sulfurylase), the two-electron reduction of APS to sulfite (APS reductase), and the six-electron reduction of sulfite to sulfide (assimilatory sulfite reductase) — plus the immediate electron-transfer architecture for the sulfite step.

Kept separate (neighboring pathways / overview maps): - **Sulfate import** (upstream): CysA/CysW/CysU + Sbp ABC transporter. - **Siroheme biosynthesis** (supporting cofactor): CysG. - **Sulfide → cysteine/methionine incorporation** (downstream): CysK, CysE/SAT, Met enzymes. - **Organosulfur scavenging** (sulfate-starvation stimulon): sulfonatases (Ssu/Msu), taurine system (Tau), arylsulfatases, sulfurtransferases. - **Broad overview map:** KEGG `ppu00920` "Sulfur metabolism" is an *overview* map, not the module; it aggregates dissimilatory, organosulfur and trafficking reactions that do not belong to the APS assimilatory branch.

Alternate names / DB definitions: "assimilatory sulfate reduction (APS route)"; KEGG module **M00176** ("Assimilatory sulfate reduction, sulfate ⇒ H2S"); MetaCyc **PWY-5340/SO4ASSIM-PWY** ("sulfate reduction I / assimilatory"). Note KEGG M00176 as drawn includes CysC/CysH (PAPS or APS variant) — curators must disambiguate the APS vs PAPS branch (see §5).

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3. Expected step model and satisfiability status

Step	Expected activity	KT2440 gene(s)	Status
1. Sulfate activation	sulfate adenylyltransferase (ATP), EC 2.7.7.4 → APS	**CysD PP_1303** + **CysNC PP_1304**	**covered**
2. APS reduction	adenylyl-sulfate reductase (thioredoxin), EC 1.8.4.10 → sulfite	**CysH PP_2328**	**covered** (direct-APS confirmed)
3. Sulfite reduction	assimilatory sulfite reductase → sulfide	**CysI PP_2371** (siroheme hemoprotein)	**covered**
3a. Electron supply (Fpr-fed CysI route)	ferredoxin–NADP+ reductase	**FprA PP_1638** (fpr-I)	**covered but GAP in metadata** (not in candidate list)
3b. Alternative (CysJ/CysI NADPH route)	NADPH sulfite reductase flavoprotein	(PP_0860 not a canonical CysJ)	**not the operational route**

Supporting cast present (outside strict boundary, confirms function): thioredoxin donors **TrxA (PP_5215)** / **TrxB (PP_0786)** for CysH; siroheme synthase **CysG (PP_3999)** for CysI; master regulator **CysB (PP_2327)** adjacent to *cysH*; regulator **FinR (PP_1637)** adjacent to *fprA*.

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4. Candidate genes and evidence (high-confidence, in-module)

- **CysD — PP_1303 (Q88NA9).** ATP sulfurylase small subunit, EC 2.7.7.4. *Evidence: direct* — transposon insertions in *cysD* dominate sulfate-assimilation-defective *P. putida* mutants (P 23794620). Caveat: locally bucketed to ppu00261; should be tagged to the APS module.
- **CysNC — PP_1304 (Q88NA8).** ATP sulfurylase large subunit (GTPase), EC 2.7.7.4; **bifunctional CysN–CysC fusion** (InterPro APS_kinase_dom IPR059117; Pfam PF01583 APS_kinase + CysN GTP-binding + SO4_adenylTrfase_lsu; 633 aa). *Evidence: direct genetics*

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+ domain analysis. **Caveat:** the fused CysC (APS kinase) domain confers PAPS-forming capacity, but this does **not** re-route assimilation through PAPS because CysH is an APS reductase (EC 1.8.4.10). The APS-kinase activity most likely serves PAPS demand for sulfotransfer/other reactions, not the reductive assimilatory flux.

- **CysH — PP_2328 (Q88KG2).** Adenosine 5'-phosphosulfate reductase, **EC 1.8.4.10**, 4Fe-4S, thioredoxin-dependent, 244 aa. *Evidence: UniProt annotation + enzyme-family diagnostics.* This is the pivotal gene fixing the module identity to the **direct APS branch** (vs *E. coli* PAPS reductase EC 1.8.4.8). Thioredoxin donors TrxA/TrxB present.
- **CysI — PP_2371 (Q88KB9).** Assimilatory sulfite reductase hemoprotein (beta), **siroheme + 4Fe-4S**, 550 aa. Terminal enzyme of the module; electron donor is ferredoxin (FprA route). *Evidence: UniProt cofactor annotation + Pseudomonad physiology* (P 23794620).
- **FprA — PP_1638 (Q88MD5, fpr-I).** Ferredoxin–NADP+ reductase; FinR-regulated; supplies electrons to CysI. *Evidence: direct — fprA* is necessary for effective sulfate assimilation and its overexpression complements the *finR* mutant sulfur phenotype (P 23794620). **Not currently in the candidate list.**

Supporting/regulatory (record for context): CysB (PP_2327), FinR (PP_1637), CysG (PP_3999), TrxA (PP_5215), TrxB (PP_0786). **Upstream transport (separate module):** CysA (PP_5168), CysW (PP_5169), CysU/CysT (PP_5170), Sbp (PP_5171, PP_4305). **Downstream C/Met synthesis (separate):** CysK (PP_4571), CysE/SAT paralogs (PP_0840, PP_0228, PP_1110, PP_3136), MetB (PP_0659), MetZ (PP_2001), MetXS (PP_5097). **Tangential:** CysQ/PAP phosphatase (PP_0261) recycles PAP (relevant to PAPS/sulfotransfer housekeeping, not a core reductive step).

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5. Gaps, ambiguities, and likely over-annotations

- **GAP — FprA (PP_1638) absent from candidate metadata.** The module explicitly lists an "Fpr electron supply" role; PP_1638 fills it and must be added. Paralog PP_4646 (fpr-II) is a lower-priority backup.

- **Over-annotation — PP_1703 (Q88M71) "assimilatory nitrate reductase/sulfite reductase" (EC 1.7.99.4).** Domain architecture is a **molybdopterin oxidoreductase (nitrate reductase, NapA/Nas-like)** fused to a CysJ-like FAD/NAD diaphorase, 1341 aa, **no siroheme**. It cannot perform assimilatory sulfite reduction (which requires siroheme). **Recommendation: strip the "sulfite reductase" label; reassign to nitrogen (nitrate/nitrite) assimilation.** Do **not** count toward module satisfiability.
- **Ambiguous — PP_0860 (Q88PJ0) "sulfite reductase flavoprotein component" (putative CysJ).** 849 aa flavoprotein carrying a **PepSY-associated transmembrane domain (PF03929)** on an FNR-like FAD/NAD module — not the canonical soluble *E. coli* CysJ (599 aa, no TM). Given that *P. putida* relies on FprA/ferredoxin for sulfite reduction (P 23794620), PP_0860 is unlikely to be the operational sulfite-reductase flavoprotein. **candidate_uncertain**; promote to full review.
- **No canonical CysJ in the genome (decisive).** A proteome-wide UniProt search of KT2440 returns **zero genes named *cysJ***; the only "sulfite reductase flavoprotein" is PP_0860 (non-canonical, above) and the only EC 1.8.1.2 (NADPH sulfite reductase) entry is the mis-annotated PP_1703. The absence of a bona fide CysJ NADPH-diaphorase means the **CysJ/CysI NADPH complex does not exist in KT2440**, leaving the ferredoxin/FprA-fed CysI route as the only viable sulfite-reduction architecture.
- **CysNC PAPS ambiguity.** The CysC (APS-kinase) domain means KEGG M00176's "PAPS" nodes may spuriously appear covered. Curators should record that assimilation is APS-direct (CysH EC 1.8.4.10) despite PAPS-forming capability.
- **Boundary over-inclusion (~30 genes).** Organosulfur scavenging (SsuABCDE PP_0236–0240 + paralogs PP_2764/2765, PP_3217/3219/3228/3229/3528/3553/3554; TauABCD PP_0209/0230–0233; sulfonate-binding proteins; 3-methylmercaptopropionyl-CoA dehydrogenases PP_0368/0370/2048/3554 [DMSP/methanethiol catabolism]; AMP-binding PP_2795/3553), sulfurtransferases (GlpE PP_0398, RhdA PP_4907, SseA PP_5118), sulfide:quinone oxidoreductase (PP_0053), cytochrome c (PP_3822), SoxYZ-like (PP_2677) — all belong to **sulfate-starvation/organosulfur or sulfur-trafficking** processes, not APS reduction (P 10482527, P 8800815). Keep out of module.

6. Module and GO-curation recommendations

Per-step status labels: - Step 1 (sulfate activation) → **covered** (CysD PP_1303 + CysNC PP_1304). - Step 2 (APS reduction) → **covered** (CysH PP_2328, EC 1.8.4.10). - Step 3 (sulfite reduction) → **covered** via **Ferredoxin/Fpr-fed CysI route** (CysI PP_2371 + FprA PP_1638). - Step 3, CysJ/CysI NADPH alternative → **not_expected_in_target_taxon** (genome has **no gene named *cysJ*** and no canonical soluble CysJ flavoprotein; physiology is Fpr/ferredoxin-dependent). - Electron-supply sub-role → **module_needs_revision**: add FprA (PP_1638) to the candidate set.

Module boundary: current generic boundary is correct in intent; the *local bucket* (KEGG overview map) is too broad. Recommend restricting the module gene set to CysDN, CysH, CysI (+FprA), and explicitly excluding transport, siroheme synthesis, cysteine/methionine synthesis, and organosulfur scavenging.

GO annotations to assert (with evidence codes): - CysH PP_2328 → GO:0033741 *adenylyl-sulfate reductase (thioredoxin) activity* (from EC 1.8.4.10). - CysI PP_2371 → GO:0009337/ GO:0004783-type *sulfite reductase (ferredoxin) activity* (GO:0050311) rather than NADPH (GO:0004783), consistent with the Fpr-fed route. - FprA PP_1638 → GO:0004324 *ferredoxin-NADP+ reductase activity*; annotate involvement in GO:0000103 *assimilatory sulfate reduction* (IMP,

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). - **GO-term request check:** if the module ontology lacks a distinct "sulfite reductase (ferredoxin), assimilatory, Fpr-fed" node, a term/relationship request may be warranted to separate it from the NADPH CysJ/CysI complex. - Remove/repair PP_1703 sulfite-reductase GO terms (reassign to nitrate/nitrite reductase, GO:0008940/GO:0042128).

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7. Genes to promote to full `fetch-gene` review

1. **FprA PP_1638 (fpr-I)** — add as the module electron-supply gene (highest priority; currently missing).
2. **CysH PP_2328** — confirm EC 1.8.4.10 vs 1.8.4.8 and thioredoxin dependence; anchors module identity.
3. **CysI PP_2371** — confirm ferredoxin (not NADPH) electron acceptor; terminal enzyme.

4. **CysNC PP_1304** — document the CysN–CysC fusion and clarify PAPS vs APS routing.
5. **PP_0860** — resolve whether it is a functional CysJ or a mis-labelled membrane flavoprotein (candidate_uncertain).
6. **PP_1703** — correct over-propagated "sulfite reductase" annotation → nitrate reductase.

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8. Key references

- Lewis TA, Glassing A, Harper J, Franklin MJ. *Role for ferredoxin:NAD(P)H oxidoreductase (FprA) in sulfate assimilation and siderophore biosynthesis in Pseudomonads.*

  P 23794620

  (2013). *Direct P. putida evidence:* cysD/cysNC/cysG essential; fprA required; finR = PP1637; complementation by fprA or E. coli cysJI.
- Vermeij P, Kertesz MA. *Pathways of assimilative sulfur metabolism in Pseudomonas putida.*

  P 10482527

  (1999). Cysteine/methionine routing; ssuD required for organosulfur (methanesulfonate) use.
- Beil S, Kertesz MA, Leisinger T, Cook AM. *The assimilation of sulfur from multiple sources and its correlation with expression of the sulfate-starvation-induced stimulon in Pseudomonas putida S-313.*

  P 8800815

  (1996). Organosulfur scavenging is a sulfate-limitation response.
- Bick JA, et al. *Regulation of the plant-type 5'-adenylyl sulfate reductase by oxidative stress.*

  P 11467967

  (2001). APS-reductase (EC 1.8.4.x) family reference for the direct-APS branch.
- UniProt/InterPro (2026): Q88KG2 (CysH), Q88NA9 (CysD), Q88NA8 (CysNC), Q88KB9 (CysI), Q88MD5 (FprA/PP_1638), Q88M71 (PP_1703), Q88PJ0 (PP_0860), Q88FT3 (CysG), Q88KG3 (CysB) — annotation/domain evidence.

Uncertainty & species-transfer notes

- **Direct (KT2440/P. putida) experimental:** essentiality of cysD, cysNC, cysG and requirement of fprA for sulfate assimilation (P 23794620); organosulfur scavenging physiology (PMIDs 10482527, 8800815).
- **Homology/annotation-based (this review):** CysH EC assignment (direct-APS), CysNC fusion, PP_1703 mis-annotation, PP_0860 non-canonical CysJ, FprA=PP_1638 locus assignment (inferred from finR=PP1637 adjacency).
- **Open questions for experts:** Does CysI in KT2440 accept electrons exclusively from ferredoxin, or can PP_0860 act as an NADPH flavoprotein partner under some conditions? Is fpr-II (PP_4646) redundant with fpr-I for sulfite reduction? Is APS-kinase (CysC domain of CysNC) ever used for assimilation, or solely for PAPS/sulfotransfer?
